

L2.5 Curve Sketching

§3.5 — precalc + the derivatives table

1. **A cubic, already factored.** $y = (x - 2)^2(x + 4)$

Run **DIAS**. Mark the *type* of each zero. Then expand, build the derivatives table on one number line, and sketch.

Stop. What goes in the **A** row for a polynomial? It is not blank.

2. Now a radical. $f(x) = \sqrt[3]{x}(x-4)$

Run **DIAS**, then build the **derivatives table**, then sketch.

Write f in powers of x before you differentiate. Report every row, including the ones whose answer is “none” — a test that fails still gets reported.

3. **Look at it first.** $y = 1 + \frac{2}{x} + \frac{1}{x^2}$

Sketch it. *Before* you differentiate anything, write one sentence about what this function can and cannot do.

4. **A rational with a slant asymptote.** $y = \frac{x^2 - 2x + 4}{x - 2}$

Full DIAS, then the table, then the sketch — with the slant asymptote drawn *on* the same axes, not beside them.

Stop. Long division first. What is left over, and where does it go?

5. **BOARDS** One rational per group. Sketch it completely on your wall.

Group 1 $y = \frac{1}{(x-3)^2(x+2)^2}$

Group 2 $y = \frac{3x^2 - 8x - 3}{x^3 - x}$

Group 3 $y = \frac{(x+2)^2(x-5)}{(x-3)(x+1)(x+4)}$

Group 4 $y = \frac{2x^3 - 5x^2 - 11x - 4}{3x^3 + 11x^2 + 5x - 3}$

Factor before you decide anything. When every wall is up, compare them: each of these four has a feature the other three do not.